

Zhifei Yang

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github
Beijing, China

EDUCATION

- **Peking University** Sep 2023 - Jun 2027 (Expected)
B.Sc. in Geographic Information Science (Major), Intelligent Science (Minor)
◦ GPA: 3.59 / 4.00 (rank: 20 %)
Beijing, China

AREAS OF EXPERIENCE

- **3D Vision:** Hands-on experience in 3D reconstruction and generation, including NeRF and Gaussian Splatting
- **Generative Models:** Experience in generative modeling, with a particular focus on native 3D generation
- **Large Language Models:** Basic knowledge of LLM pre-training and post-training, including SFT, RL, and on-policy distillation (OPD)
- **Character Animation & Motion Control:** Foundational knowledge of skeletal animation, kinematics, and motion control

RESEARCH EXPERIENCE

- **Beijing Institute for General Artificial Intelligence (BIGAI)** Mar 2025 - Present
Research Assistant
◦ Advisor: Siyuan Huang, Yixing Chen
◦ Research on 3D reconstruction and scene understanding
Beijing, China

INTERNSHIP EXPERIENCE

- **ByteDance** Jun 2026 - Present
3D Generative Modeling Intern
◦ Worked on large-scale 3D data curation and the training of native 3D generative models

PUBLICATIONS

- [C.1] Junfeng Ni, Yixin Chen, **Zhifei Yang**, Yu Liu, Ruijie Lu, Song-Chun Zhu, Siyuan Huang. (2026). **G4Splat: Geometry-Guided Gaussian Splatting with Generative Prior**. In *International Conference on Learning Representations (ICLR 2026)*.
- [P.1] Junfeng Ni, Zirui Zhou, Yixin Chen (corresponding author), Yu Liu, Nan Jiang, **Zhifei Yang**, Songchun Zhu, Siyuan Huang. (2026). **DecomVoxel: Harnessing 3D-Native Priors with Guided In-situ Denoising Optimization for Compositional Scene Reconstruction**. Manuscript.

PROJECTS

- **G4Splat: Geometry-Guided Gaussian Splatting with Generative Prior** Mar 2025 - Sep 2025
@DaLi-Jack / G4Splat
 - Combined geometric priors and generative models to enhance Gaussian Splatting under large viewpoint changes and sparse inputs, **published at ICLR 2026**
 - Extracted 3D planes from multi-view images and developed a visibility-aware mechanism to guide optimization toward unseen regions
 - Reconciled multi-view geometry with generative priors, achieving over 20% improvement on multiple benchmarks
- **Native 3D Generation with Explicit Topology** Jun 2026 - Present
 - Developing a native 3D generative framework that jointly models vertex geometry and mesh connectivity, targeting meshes with plausible topology rather than surface geometry alone
 - Training VAE and flow-matching models for structured mesh representation learning and generation
 - Building the data-processing pipeline and curating tens of millions of 3D assets for large-scale training

HONORS & AWARDS

- **Peking University Third Prize Scholarship (Top 20%)** Jan 2025
Peking University